

Solution: Linear Systems from CIE-1

Problem Analysis

The problem asks us to relate the solvability of a linear system $Ax = b$ to the geometric arrangement of the column vectors of matrix A (denoted as a_1 and a_2) and the target vector b .

The system of linear equations can be written as a vector equation:

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b}$$

This means we are looking for weights x_1 and x_2 such that the linear combination of vectors a_1 and a_2 equals vector b .

Visual Analysis of the Figures

1. Fig Q1 (i):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line passing through the origin. They are **collinear** (linearly dependent).
- Their span (the set of all possible linear combinations) is just that single line.
- Vector $\mathbf{b}$ points in a different direction and does **not** lie on that line.

2. Fig Q1 (ii):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ point in different directions. They are **non-collinear** (linearly independent).
- Because they are two linearly independent vectors in 2D space ($\mathbb{R}^2$), they form a basis. Their span covers the entire 2D plane.
- Vector $\mathbf{b}$ lies somewhere in that plane (as any 2D vector must).

3. Fig Q1 (iii):

- Vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ lie on the same line. They are **collinear** (linearly dependent).
- Vector $\mathbf{b}$ **also** lies on exactly the same line.

Detailed Solutions (Question 1)

(a) Unique Solution

Correct Figure: Fig Q1 (ii)

Mathematical Reasoning: A system $Ax = b$ has a **unique solution** if and only if the columns of A are linearly independent. In $\mathbb{R}^2$, this means the two column vectors must not lie on the same line. In Fig Q1 (ii), $\mathbf{a}_1$ and $\mathbf{a}_2$ span the entire plane, meaning any vector $\mathbf{b}$ can be reached by a unique combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Exam-Style Justification (Max 2 sentences): In Fig Q1 (ii), the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are linearly independent (non-collinear), meaning they span the entire plane $\mathbb{R}^2$. Therefore, any vector $\mathbf{b}$ in the plane can be represented uniquely as a linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

(b) No Solution

Correct Figure: Fig Q1 (i)

Mathematical Reasoning: A system has **no solution** (is inconsistent) if the target vector $\mathbf{b}$ lies outside the column space (span) of $\mathbf{A}$. In Fig Q1 (i), the span of $\mathbf{a}_1$ and $\mathbf{a}_2$ is a single line. Since $\mathbf{b}$ is off that line, there are no scalar multipliers x_1, x_2 that can make $x_1\mathbf{a}_1 + x_2\mathbf{a}_2$ equal to $\mathbf{b}$.

Exam-Style Justification (Max 2 sentences): In Fig Q1 (i), the vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ are collinear, so their span is limited to a single line. Since vector $\mathbf{b}$ does not lie on this line, it cannot be generated by any linear combination of $\mathbf{a}_1$ and $\mathbf{a}_2$.

Summary for Reference

- **Fig Q1 (i):** No Solution (inconsistent).
- **Fig Q1 (ii):** Unique Solution (consistent and independent).
- **Fig Q1 (iii):** Infinitely Many Solutions (consistent but dependent; $\mathbf{b}$ is on the line spanned by $\mathbf{a}_1, \mathbf{a}_2$).

Solution: Vector Spaces of Matrices (Question 2)

Problem Statement

Identify which of the following sets is **not** a vector space over $\mathbb{R}$: (i) $\mathcal{M}_{2 \times 2}$, the set of real, invertible 2×2 matrices. (ii) $\mathcal{U}_{3 \times 3}$, the set of real, upper triangular matrices.

Detailed Analysis

(i) Set of Invertible 2×2 Matrices ($\mathcal{M}_{2 \times 2}$)

Conclusion: This is **NOT** a vector space.

Reasoning: A vector space must satisfy several axioms, including the existence of an additive identity (the zero vector) and closure under addition.

1. **Missing Zero Vector:** The additive identity in the space of 2×2 matrices is the zero matrix $O = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. The determinant of the zero matrix is 0, so it is **not invertible**. Therefore, the zero vector is not included in the set.
2. **Not Closed under Addition:** Even if we ignored the zero vector issue, the sum of two invertible matrices is not guaranteed to be invertible.

- Let $A = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (Invertible, $\det = 1$).

- Let $B = -I = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ (Invertible, $\det = 1$).
- $A + B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, which is **not** invertible.

Since the set fails to contain the zero vector and is not closed under addition, it is not a vector space.

(ii) Set of Upper Triangular Matrices ($\mathcal{U}_{3 \times 3}$)

Conclusion: This **IS** a vector space.

Reasoning: This set is a subspace of the vector space of all 3×3 real matrices, $M_{3 \times 3}(\mathbb{R})$.

1. **Contains Zero Vector:** The zero matrix consists entirely of zeros. Since all entries below the main diagonal are zero ($0 = 0$), it satisfies the condition of being upper triangular.
2. **Closed under Addition:** If you add two upper triangular matrices, the entries below the diagonal remain zero ($0 + 0 = 0$).
3. **Closed under Scalar Multiplication:** If you multiply an upper triangular matrix by a scalar c , the entries below the diagonal remain zero ($c \times 0 = 0$).

Final Answer

The set that is **not a vector space** is (i) $\mathcal{M}_{2 \times 2}$, the set of real, invertible 2×2 matrices. The primary reason is that it does not contain the zero matrix (additive identity), and it is not closed under vector addition.

Solution: Assertions on Vector Spaces (Question 3)

Problem Statement

Assert or Reject each of the following 3 statements with proper justification.

1. **Statement 1:** The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set.
2. **Statement 2:** The set containing a single vector is **ALWAYS** a basis for some vector subspace.
3. **Statement 3:** Every vector space has at least one subspace.

Detailed Analysis

Statement 1: "The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set."

Verdict: REJECT (False)

Justification: A set containing a single vector $\{\mathbf{v}\}$ is linearly dependent if and only if that vector is the zero vector ($\mathbf{v} = \mathbf{0}$). If the vector is non-zero ($\mathbf{v} \neq \mathbf{0}$), the set is linearly **independent**. This is because the equation $c\mathbf{v} = \mathbf{0}$ implies $c = 0$ when $\mathbf{v}$ is not zero.

- **Counter-example:** Let $V = \mathbb{R}^2$ and let $S = \{(1, 0)\}$. Since $(1, 0) \neq \mathbf{0}$, the set S is linearly independent. Thus, the statement "ALWAYS dependent" is false.

Statement 2: "The set containing a single vector is ALWAYS a basis for some vector subspace."

Verdict: REJECT (False)

Justification: By definition, a basis for a subspace must be a **linearly independent** spanning set. If the single vector in the set is the zero vector, $S = \{\mathbf{0}\}$, then the set is linearly **dependent** (since $1 \cdot \mathbf{0} = \mathbf{0}$, a non-trivial combination yields zero). Therefore, the set $\{\mathbf{0}\}$ cannot be a basis for any subspace. (Note: The subspace generated by the zero vector is the zero subspace $\{\mathbf{0}\}$, but the basis for the zero subspace is conventionally defined as the empty set $\emptyset$, not the set containing the zero vector.)

Statement 3: "Every vector space has at least one subspace."

Verdict: ASSERT (True)

Justification: Every vector space V must contain the additive identity (the zero vector, $\mathbf{0}$). The set $W = \{\mathbf{0}\}$ is a subspace of V because it contains the zero vector and is closed under addition ($\mathbf{0} + \mathbf{0} = \mathbf{0}$) and scalar multiplication ($c \cdot \mathbf{0} = \mathbf{0}$). This is known as the **zero subspace** (or trivial subspace). Therefore, every vector space has at least this one subspace. (Most vector spaces actually have at least two: the zero subspace $\{\mathbf{0}\}$ and the space V itself.)

Solution: Multiple Solutions to a Linear System (Question 4)

Problem Statement

Let A be a 3×3 matrix such that $Ax = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ has both $\mathbf{x}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$ as its solutions. Find one more solution for the same.

Detailed Solution

1. Understanding the Linear System Structure

We are given a non-homogeneous linear system $Ax = b$. The general solution to such a system is of the form:

$$x = x_p + x_h$$

where:

- x_p is a particular solution to $Ax = b$.
- x_h is any solution to the homogeneous system $Ax = 0$ (i.e., x_h is in the null space of A).

2. Finding a Homogeneous Solution

Since both $\mathbf{x}_1$ and $\mathbf{x}_2$ are solutions to $A\mathbf{x} = \mathbf{b}$, their difference must be a solution to $A\mathbf{x} = \mathbf{0}$.

$$A(\mathbf{x}_1 - \mathbf{x}_2) = A\mathbf{x}_1 - A\mathbf{x}_2 = \mathbf{b} - \mathbf{b} = \mathbf{0}$$

Let's calculate this difference vector, let's call it $\mathbf{v}$:

$$\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix}$$

So, $A \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}$.

3. Generating a New Solution

We can create a new solution by taking one of the original solutions (like $\mathbf{x}_1$) and adding any scalar multiple of the homogeneous solution $\mathbf{v}$. New Solution $\mathbf{x}_3 = \mathbf{x}_1 + k\mathbf{v}$ (where k is any real number).

Let's choose $k = 1$ for simplicity:

$$\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

Verification

Since $\mathbf{x}_3 = \mathbf{x}_1 + \mathbf{v}$:

$$A\mathbf{x}_3 = A(\mathbf{x}_1 + \mathbf{v}) = A\mathbf{x}_1 + A\mathbf{v} = \mathbf{b} + \mathbf{0} = \mathbf{b}$$

This confirms $\mathbf{x}_3$ is a valid solution.

Alternative Approach (Affine Combination)

Any affine combination of solutions is also a solution (where weights sum to 1).

$\mathbf{x}_3 = 2\mathbf{x}_1 - 1\mathbf{x}_2$ (This matches the calculation above). Another option, $\mathbf{x}_4 = 2\mathbf{x}_2 - \mathbf{x}_1$:

$$2 \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 8 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$$

Final Answer

One possible additional solution is:

$$\mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$$

(Note: There are infinitely many correct answers. Any vector on the line connecting the two given points is a solution.)

Solution: Fundamental Subspaces (Question 5)

Problem Statement

Given matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}$, find: (i) The four fundamental subspaces. (ii) A basis for each. (iii)

The dimension of each.

Detailed Analysis

First, we determine the rank of the matrix. The columns are identical, so there is only 1 independent column. **Rank (r) = 1**. Dimensions of matrix A are $m \times n = 3 \times 2$.

- $m = 3$ (rows)
- $n = 2$ (columns)

1. Column Space ($C(A)$)

- **Definition:** The space spanned by the columns of A .
- **Analysis:** Since both columns are the same $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$, the span is just the line generated by this single vector.
- **Basis:** $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $r = 1$

2. Row Space ($C(A^T)$)

- **Definition:** The space spanned by the rows of A .
- **Analysis:** All rows are identical $(1, 1)$. The row space is the line in $\mathbb{R}^2$ generated by this vector.
- **Basis:** $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $r = 1$

3. Null Space ($N(A)$)

- **Definition:** The set of vectors x such that $Ax = 0$.

- **Calculation:** $\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ This reduces to the single equation $x_1 + x_2 = 0$, or $x_1 = -x_2$. Vector solution: $x = x_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix}$.
- **Basis:** $\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $n - r = 2 - 1 = 1$

4. Left Null Space ($N(A^T)$)

- **Definition:** The set of vectors y such that $A^T y = 0$.
- **Calculation:** $A^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$ Solve $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. This gives one equation: $y_1 + y_2 + y_3 = 0$. We have 2 free variables (y_2, y_3). Let $y_1 = -y_2 - y_3$. Vector solution: $y = \begin{pmatrix} -y_2 - y_3 \\ y_2 \\ y_3 \end{pmatrix} = y_2 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + y_3 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.
- **Basis:** $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$
- **Dimension:** $m - r = 3 - 1 = 2$

Summary Table

Subspace	Symbol	Dimension	Basis
Column Space	$C(A)$	1	$\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$
Row Space	$C(A^T)$	1	$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$
Null Space	$N(A)$	1	$\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$
Left Null Space	$N(A^T)$	2	$\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$

Solution: Zero Vector in Non-Standard Space (Question 6)

Problem Statement

Find the zero vector Θ_2 for the set $\mathbb{R}^2$ with the following operations: **Vector Addition ($\oplus$)**:

$$\mathbf{x} \oplus \mathbf{y} = \begin{pmatrix} x_1 + y_1 - 2 \\ x_2 + y_2 - 3 \end{pmatrix}$$

Scalar Multiplication ($\odot$):

$$k \odot \mathbf{x} = \begin{pmatrix} kx_1 - 2k + 2 \\ kx_2 + 3k - 3 \end{pmatrix}$$

Detailed Solution

1. Definition of the Zero Vector

The zero vector (additive identity), denoted here as Θ_2 , is the vector such that for any vector $\mathbf{x} \in \mathbb{R}^2$, the following condition holds:

$$\mathbf{x} \oplus \Theta_2 = \mathbf{x}$$

2. Setting up the Equations

Let $\Theta_2 = \begin{pmatrix} e_1 \\ e_2 \end{pmatrix}$. Substitute $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ and Θ_2 into the given addition rule:

$$\mathbf{x} \oplus \Theta_2 = \begin{pmatrix} x_1 + e_1 - 2 \\ x_2 + e_2 - 3 \end{pmatrix}$$

For this result to equal $\mathbf{x}$, each component must match the components of $\mathbf{x}$:

$$\begin{pmatrix} x_1 + e_1 - 2 \\ x_2 + e_2 - 3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

3. Solving for Components

We solve for e_1 and e_2 individually:

Row 1:

$$x_1 + e_1 - 2 = x_1$$

$$e_1 - 2 = 0 \implies e_1 = 2$$

Row 2:

$$x_2 + e_2 - 3 = x_2$$

$$e_2 - 3 = 0 \implies e_2 = 3$$

Final Answer

The zero vector Θ_2 for this specific vector space is:

$$\Theta_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Solution: Null Space of a Single Equation (Question 7)

Problem Statement

Consider the following system of equation:

$$x_1 + x_2 - x_3 + x_4 - x_5 = 0$$

Identify the null space associated with the coefficient matrix and its dimension.

Detailed Analysis

1. Identify the Coefficient Matrix (A)

The system consists of a single homogeneous linear equation with 5 variables. The coefficient matrix A is a 1×5 matrix:

$$A = \begin{pmatrix} 1 & 1 & -1 & 1 & -1 \end{pmatrix}$$

We are looking for the Null Space $N(A)$, which is the set of all vectors $\mathbf{x} \in \mathbb{R}^5$ such that $A\mathbf{x} = 0$.

2. Determine Rank and Nullity

- **Number of Variables (n):** 5
- **Rank (r):** The matrix has 1 pivot (the first entry, 1). Thus, $r = 1$.
- **Nullity (Dimension of Null Space):** By the Rank-Nullity Theorem:

$$\text{Nullity} = n - r = 5 - 1 = 4$$

3. Find the Basis for the Null Space

To find the null space, we express the general solution in terms of the free variables.

- **Pivot Variable:** x_1
- **Free Variables:** x_2, x_3, x_4, x_5

Rewrite the equation to solve for the pivot variable:

$$x_1 = -x_2 + x_3 - x_4 + x_5$$

Substitute this into the general vector $\mathbf{x}$:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} -x_2 + x_3 - x_4 + x_5 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$$

Decompose this vector into a linear combination of vectors weighted by the free variables:

$$\mathbf{x} = x_2 \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} + x_5 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

4. Final Answer

- **Null Space:** The null space is the set of all linear combinations (the span) of the four basis vectors found above.

$$N(A) = \text{span} \left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

- **Dimension:**

$$\text{Dimension} = 4$$

Solution: Geometry of Subspaces in R3 (Question 8)

Problem Statement

Given the vector space $\mathbb{R}^3$, list out the geometry of its possible subspaces.

Detailed Analysis

Subspaces of $\mathbb{R}^3$ are categorized by their dimension. Since we are working in a 3-dimensional space, the possible dimensions of subspaces range from 0 to 3. Every subspace must pass through the origin (contains the zero vector).

1. Dimension 0: The Origin

- **Algebraic Description:** The set containing only the zero vector $\{\mathbf{0}\}$.
- **Geometry:** A single **point** at the origin $(0, 0, 0)$.
- **Basis:** The empty set $\emptyset$ (or sometimes conventionally denoted as containing the zero vector, though technically independent sets cannot contain zero).

2. Dimension 1: Lines through the Origin

- **Algebraic Description:** The span of a single non-zero vector $\mathbf{v}$. $S = \{c\mathbf{v} \mid c \in \mathbb{R}\}$
- **Geometry:** A **line** passing through the origin.
- **Basis:** Any single non-zero vector lying on that line.

3. Dimension 2: Planes through the Origin

- **Algebraic Description:** The span of two linearly independent vectors $\mathbf{u}$ and $\mathbf{v}$.
 $S = \{c_1\mathbf{u} + c_2\mathbf{v} \mid c_1, c_2 \in \mathbb{R}\}$
- **Geometry:** A **plane** passing through the origin.
- **Basis:** Any two linearly independent vectors lying on that plane.

4. Dimension 3: The Entire Space

- **Algebraic Description:** The entire set $\mathbb{R}^3$.
- **Geometry:** The whole 3D **space** itself.
- **Basis:** Any three linearly independent vectors (e.g., standard basis $\mathbf{i}, \mathbf{j}, \mathbf{k}$).

Summary Table

Dimension	Geometric Object	Restriction
0	Point	Must be the Origin $(0, 0, 0)$